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SPORTS PERFORMANCE



Elite versus non-elite cyclist – Stepping up to the international/elite ranks from U23 cycling

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ABSTRACT

This study investigated the physiological, performance and training characteristics of U23 cyclists and assessed the requirements of stepping up to the elite/international ranks. Twenty highly trained U23 cyclists (age, 22.1 ± 0.8 years; body mass, 69.1 ± 6.8 kg; $\text{VO}_{2\text{max}}$, 76.1 ± 3.9 ml·kg⁻¹·min⁻¹) participated in this study. The cyclists were *a posteriori* divided into two groups based on whether or not they stepped up to elite/international level cycling (U23_{ELITE} vs. U23_{NON-ELITE}). Physiological, performance and training and racing characteristics were determined and compared between groups. U23_{ELITE} demonstrated higher absolute peak power output ($p = .016$), 2 min ($p = .026$) 5 min ($p = .042$) and 12 min ($p \leq .001$) power output as well as higher absolute critical power ($p = .002$). Further, U23_{ELITE} recorded more accumulated hours ($p \leq .001$), covered distance ($p \leq .001$), climbing metres ($p \leq .001$), total sessions ($p \leq .001$), total work ($p \leq .001$) and scored more UCI points ($p \leq .001$). These findings indicate that U23_{ELITE} substantially differed from U23_{NON-ELITE} regarding physiological, performance and training and racing characteristics derived from laboratory and field. These variables should be considered by practitioners supporting young cyclists throughout their development towards the elite/international ranks.

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KEYWORDS

Endurance performance; maturation; high performance; talent identification

Introduction

The international cycling governing body (Union Cycliste Internationale, UCI) specifies three age-related categories, i.e. juniors (17 to 18 years), U23 (19 to 22 years) and elite (> 23 years). Development and maturation of performance capabilities, an appropriate training strategy and the accumulation of race experience play a pivotal role in the long-term development of competitive road cyclists throughout these age categories (Leo, Spragg, Simon, Mujika et al., 2021; Menaspà et al., 2012; Mostaert et al., 2021; Nimmerichter, 2018).

The physiological and performance attributes of junior and professional road cyclists are consistent within the literature (Jeukendrup et al., 2000; Menaspà et al., 2012; Mujika & Padilla, 2001; Pinot & Grappe, 2015; Sallet et al., 2006), while those of U23 cyclists remain conflicting and subject to debate (Leo et al., 2022; Leo, Spragg, Menz et al., 2021; Leo, Spragg, Simon, Lawley et al., 2021; Leo, Spragg, Simon, Mujika et al., 2021). However, numerous studies showed that late junior and U23 race success may be considered as predictor of future success (i.e. stepping up to the elite/international level; Mostaert et al., 2021; Svendsen et al., 2018). For example, Mostaert et al., (2021) demonstrated that superior race success (i.e. reaching a top 10 finish) at age 17 to 19 provided a higher chance to reach elite/international level, and Svendsen et

al.,(2018) found that those cyclists who made it to the elite/international level by age 23 already demonstrated higher annual racing hours in the junior category (age 18) than their age related counterparts. Thus, the U23 cycling category represents an important transition in the making of a future elite/international cyclist (Leo, Spragg, Simon, Mujika et al., 2021; Svendsen et al., 2018), but little research has focused on the performance determinants, race demands and training characteristics of this category (Leo et al., 2022; Leo, Spragg, Menz et al., 2021; Leo et al., 2020; Leo, Spragg, Simon, Mujika et al., 2021). In addition, the majority of the available research (Bell et al., 2017; Coyle, 2005; Coyle, 1995; Jeukendrup et al., 2000; Lucia et al., 2001) focused on lab-derived measures such as peak power output (PPO), maximum oxygen uptake ($\text{VO}_{2\text{max}}$) or fractional utilization of $\text{VO}_{2\text{max}}$ (% $\text{VO}_{2\text{max}}$), while only one study reported field derived measures (Pinot & Grappe, 2014). Although laboratory based testing provides unique and fundamental information about the physiological capacity of an athlete, field based methods (i.e. performance testing and training monitoring) could assist to refine athlete profiling (Gallo et al., 2022; Leo, Spragg, Menz et al., 2021; Nimmerichter et al., 2010; Podlogar et al., 2022). A cyclist's power profile, training characteristics

and race performances are strong determinants of the talent identification process (Galloet al., 2022; Leo, Spragg, Simon, Mujika et al., 2021). More recently the decline in the power profile after accumulated work (i.e. >2000 kJ) has appeared as a blueprint of elite/international cycling performance (Gallo et al., 2022; Leo, Spragg, Simon, Mujika et al., 2021; Mateo-March et al., 2022; Muriel, Mateo-March et al., 2021; Spragg et al., 2022; Van Erp, Sanders et al., 2021).

In this context, the paucity of information about the physiological, performance and training requirements of U23 cycling does not allow to assess the determinants for riders in this age group to step up to the elite/international ranks. Therefore, the aim of this study was to investigate the characteristics of U23 cyclists who stepped up to UCI professional teams (U23_{ELITE}) vs. those who did not (U23_{NON-ELITE}). The authors hypothesized that U23_{ELITE} cyclists demonstrate higher physiological, performance, as well as training and racing characteristics compared to U23_{NON-ELITE}.

Methods

Participants

Initially, twenty-seven highly trained (McKay et al., 2021) male U23 cyclists (age, 21.4 ± 1.2 years; height, 182.3 ± 4.9 cm; weight, 69.7 ± 6.3 kg; VO_{2max} , 75.2 ± 4.2 ml·kg⁻¹·min⁻¹) were assigned to this study during a four year time frame (2018 to 2021). At the time of participation, the subjects were active members of the same UCI licenced U23 Continental team. The team followed all anti-doping policies as set out in the world anti-doping agency code and the team was also an active member of “Mouvement Pour un Cyclisme Crédible” (movement for a credible cycling) union. Importantly, no antidoping violations were recorded for any participant during the study duration. Recruitment was based on voluntary interest. In cases of any prolonged illness, injury, or termination of cycling career, participants were excluded from all analysis. For this reason, seven participants (N = 7) were excluded from further analysis resulting in a final N of 20 subjects (N = 20, age, 22.1 ± 0.8 years; height, 181.8 ± 5.2 cm; weight, 76.1 ± 3.9 kg; VO_{2max} , 76.1 ± 3.9 ml·kg⁻¹·min⁻¹).

Informed written consent was obtained after each participant was given a verbal and written explanation of the experimental protocol and fully understood the possible risks involved in taking part in the study. The study protocol was approved by the Ethical Review Board at the University of Innsbruck and followed the principles as set out in the declaration of Helsinki.

Study design

The study design involved: 1) laboratory testing; 2) power profiling; and 3) quantification of training and racing characteristics. The cycling season was a 365-day period from 1st November to 31st October for the season analysed. The laboratory and power profiling procedures were in line with previously used methods by the authors (Leo, Spragg, Menz et

al., 2021; Leo et al., 2020). The study participants were *a posteriori* divided into two groups: 1) elite group turning professional – riders who signed a pro contract with a UCI licenced Pro or World Team (U23_{ELITE}); and 2) non-elite group – subjects who did not step up to the elite/international level (McKay et al., 2021; U23_{NON-ELITE}). The respective season analysed involved either the season before turning professional (U23_{ELITE}) or the last year in the U23 category (U23_{NON-ELITE}). In a first step all recorded data regarding physiological testing, power profiling as well as training and racing characteristics were compared between groups (U23_{ELITE} vs. U23_{NON-ELITE}). Data files containing erroneous or missing data due to failure of power metres and/or bike changes during racing were excluded from further analyses. In a second step the relationships between the scored UCI points and the key performance indicators for turning professional were investigated.

Laboratory testing

Laboratory testing was conducted in the first week of March for each respective season. All laboratory tests were completed in the same lab using the same measurement equipment. Open circuit breath by breath spirometry (ZAN600, nSpire Health GmbH, Germany) was used to record tidal volume, breathing frequency and volume percentages of oxygen (O₂) and carbon dioxide (CO₂). Volume and gas calibrations were conducted before each measurement according to the manufacturer's recommendations. Ventilation (STPS) was then derived using the Haldane transformation to calculate oxygen uptake (VO₂) and carbon dioxide (VCO₂) release (Wilmore & Costill, 1973). All tests were performed on an electromagnetically braked stationary trainer (Cyclus2, RBM elektronik-automation GmbH, Leipzig, Deutschland) with the participants' own road bikes (Alto Prestige and Revelator, KTM Fahrrad GmbH, Mattighofen, Austria). Laboratory testing involved a standardized warm-up of 3 min at 100 W, followed by a ramp incremental graded exercise test (GXT). The initial workload for the GXT was set at 150 W and was increased by 20 W·min⁻¹ until volitional exhaustion. The measurements included absolute and relative PPO and VO_{2max}. PPO and VO_{2max} were defined as the highest 30s average during the GXT (Scheidler et al., 2017). The ventilatory threshold (VT) and the respiratory compensation point (RCP) were analysed from the GXT according to Beaver et al. (1986). VT was defined as the point where the ventilation rate (VE) increased compared to VO₂ (VE/VO₂). RCP was defined as the onset of hyperventilation during the GXT, with an increase in VE compared to the volume of carbon dioxide (VCO₂) release, known as the VE/VCO₂ ratio. Both VT and RCP were independently analysed via visual inspection by two co-authors within an agreement of 10 W, otherwise a third independent expert was involved. Continuous recordings of heart rate (HR) were performed via short range telemetry with a 1 Hz sampling rate (V800, Polar Electro Oy, Kempele, Finland).

Power profiling

The power profiling field tests, consisting of 2, 5 and 12 min maximal efforts, were conducted in line with the laboratory testing within a 10-day time frame. All efforts were performed

on a standardized uphill climb with an average gradient of 5.5% on two consecutive days. After a standardized warm up, day one randomly applied the 2 and 5 min maximum efforts interspersed by 30 min of active recovery at a rating of perceived exertion (RPE) <2 out of 10. On day two the 12 min maximum effort was conducted. Prior to each effort the participants were encouraged to produce the highest possible workload at a freely chosen cadence.

Power output was recorded using a standardized crank system (SRAM Red, Quarq, Spearfish, South Dakota, USA) with a 1-Hz sampling rate. A static calibration of each power metre unit was undertaken prior to each field test according to Wooles et al. (2005). Participants were instructed to perform a “zero-offset” before each effort.

Power output values were analysed from 2, 5 and 12 min efforts using an online cycling software platform (EnDuRa, Blue Cat Technical Ltd, Midhurst, West Sussex, United Kingdom). All power metre data were derived as the highest rolling average to the corresponding duration of 2, 5 and 12 min. In addition, all data were manually checked for data spikes applying a three standard deviation threshold (Morris & Langari, 2020)

CP and W' were estimated applying the non-linear power output-time, the linear work-time and the linear power output-inverse of time model combined with non-linear or linear least square regressions analyses. The standard errors for both parameter estimates, CP and W' , were calculated for every participant and model. As recently proposed, parameter estimates from the model with the smallest sum of standard errors was used for every participant. The power output-inverse of time model (Equation 1) showed the smallest sum of standard errors (Black et al., 2015) for every participant in the current investigation, and thus, CP and W' derived from this model were used for all further analyses.

$$P = W' \times \frac{1}{TTE} + CP \quad (1)$$

Equation 1: P – power output (W); TTE – time to task failure (s); CP – critical power (W); W' – work capacity above critical power (kJ).

Additionally, 2, 5 and 12 min mean maximum power outputs (MMP) from racing after 2000 kJ of accumulated total work were analysed using an online cycling software platform (EnDuRa, Blue Cat Technical Ltd, Midhurst, West Sussex, United Kingdom). Moreover, the percentage decline (%dec) in comparison to the 2, 5 and 12 min efforts from the power profile test was calculated.

Training and racing characteristics

The accumulated training hours, covered distance, number of training sessions, climbing metres and total work for road and time trial bike activities were recorded on a day-to-day basis over an entire season (48 weeks). Heart rate (HRM-Dual, Garmin Ltd, Schaffhausen, Switzerland) and power output data were recorded at a 1-Hz sampling rate and saved on each participant's head unit (Garmin Edge 520 and 530, Garmin (Europe) Ltd., Southampton, United Kingdom).

All athletes uploaded their training and racing data to an online training platform (Trainingpeaks, Trainingpeaks LLC, Winchester, USA). All training data were further processed,

analysed, and checked for data spikes in Microsoft Excel (Excel, Microsoft Corporation, Redmond, USA). Total time and percentage time (%Time) in HR and power output intensity zones were based on the individual VT and RCP values from the laboratory GXT using a three-zone intensity distribution: 1) zone 1 < VT, 2) zone 2 VT<>RCP and 3) zone 3 > RCP. Power output from the laboratory GXT was corrected for the mean response time (MRT) in VO_2 (Boone et al., 2008; Iannetta et al., 2019). Lucia's training impulse model (Sanders et al., 2017; luTRIMP) was derived from the three-zone heart rate intensity distribution adding a zone-specific weighting factor (i.e. 1 for zone 1; 2 for zone 2, and 3 for zone 3). A polarization index was included to better interpret and evaluate intensity distribution (Treff et al., 2019).

Number of race days, race kilometres and top three race results from single stage and multi-stage races were analysed for each rider using an online cycling database (First Cycling, FirstCycling AS, Indre Arna, Norway). The total UCI points scored – in arbitrary units (A.U.) – were derived from the top 3 race results for 1) single stage events: 1. Pro, 1.1, 1.2, nation cup (NC); and 2) multi-stage events: 2. Pro, 2.1, 2.2, NC, which were based on the UCI classification guidelines (UCI).

Statistical analysis

Descriptive data are presented as mean $\pm$ standard deviation (SD) or mean difference (Δ). All recorded data were initially checked for violations to normality with a Shapiro-Wilk test. Independent t-tests, or Mann-Whitney U test when normality was violated, with Cohen's d effect size were performed for all variables derived from laboratory and field testing, MMP and % dec after 2000 kJ, as well as training and racing characteristics between U23_{ELITE} and U23_{NON-ELITE} groups. Equality of variances were controlled with Levene's test. Cohen's d effect size was set at 0.2 to 0.5 for small, 0.5 to 0.8 for moderate and >0.8 for a large effect. Spearman's rho (ρ) rank correlation coefficient investigated the relationship between the total UCI points scored and the key performance indicators, which were significantly different between groups. The correlation coefficient rho (ρ) was interpreted as small (0.1 to 0.3), moderate (0.3 to 0.5) or large (>0.5) effect according to Hopkins (2002). All statistical analyses (JASP 15.0, JASP Team, Amsterdam, the Netherlands) and graphical illustrations (GraphPad Prism 8, GraphPad Software, San Diego, USA) were conducted with commercially available software packages. The level of statistical significance was set at an α level < .050 two tailed.

Results

Laboratory testing

Participants' anthropometric and physiological characteristics from the laboratory testing are presented in Table 1.

Power profiling

Absolute 2 min (Δ = 27 W, p = .026, d = .85), 5 min (Δ = 26 W, p = .042, d = .97) and 12 min (Δ = 35 W, p $\leq$.001, d = 1.75) power output, and CP (Δ = 33 W, p = .002, d = 1.59) were significantly

Table 1. Anthropometrical and physiological measures between U23_{ELITE} and U23_{NON-ELITE}; VO_{2max} – maximum oxygen uptake; PPO – peak power output; RCP – respiratory compensation point; VT – ventilatory threshold; *significantly different from U23_{ELITE}.

	U23 _{ELITE} (n = 10)	U23 _{NON-ELITE} (n = 10)	p-value and effect size
Age (years)	21.7 ± 1.0	22.5 ± 0.2*	.027* 1.07
Body Mass (kg)	71.5 ± 7.2	67.0 ± 5.8	.172 .64
Height (cm)	184.1 ± 3.6	179.6 ± 5.7	.053 .93
VO _{2max} (L.min ⁻¹ / ml.kg ⁻¹ .min ⁻¹)	5.5 ± 0.5 76.2 ± 5.1	5.0 ± 0.5 75.1 ± 4.2	.055/.919 .292/.48
PPO (W/W.kg ⁻¹)	486 ± 38 6.8 ± 0.4	443 ± 33* 6.6 ± 0.5	.016/1.19* .380/.40
RCP (W/W.kg ⁻¹)	382 ± 32 5.4 ± 0.3	356 ± 35 5.3 ± 0.3	.983/.78 .648/.21
VT (W/W.kg ⁻¹)	253 ± 22 3.5 ± 0.3	238 ± 22 3.5 ± 0.3	.118/.73 .983/.10

higher in U23_{ELITE} than in U23_{NON-ELITE} – see, Figure 1A. However, no significant differences were found in 2 min ($\Delta = 0.1 \text{ W.kg}^{-1}$, $p = .629$, $d = .21$), 5 min ($\Delta = 0.1 \text{ W.kg}^{-1}$, $p = .820$, $d = .10$), and 12 min ($\Delta = 0.1 \text{ W.kg}$, $p = .606$, $d = .24$) power output, nor in CP ($\Delta = 0.1 \text{ W.kg}^{-1}$, $p = .490$, $d = .32$) when normalized to body mass. No significant differences between groups were found in W' derived from the power profiling test ($\Delta = 0.8 \text{ kJ}$, $p = .564$, $d = .26$) – see, Table 2.

Additionally, absolute 2 min ($\Delta = 66 \text{ W}$, $p = .001$, $d = 1.69$), 5 min ($\Delta = 50 \text{ W}$, $p = .003$, $d = 1.56$) and 12 min ($\Delta = 56 \text{ W}$, $p \leq .001$, $d = 2.38$) MMP values after 2000 kJ were higher in U23_{ELITE} vs. U23_{NON-ELITE} – see, Figure 1B. The %dec between 2 min ($\Delta = 7.7\%$, $p = .001$, $d = 1.74$), 5 min ($\Delta = 5.7\%$, $p \leq .001$, $d = 1.98$) and 12 min ($\Delta = 5.6\%$, $p = .001$, $d = 1.69$) power output during the power profile test, and the respective MMP values after 2000 kJ were lower in U23_{ELITE} than in U23_{NON-ELITE} – see, Figure 1C.

Training and racing characteristics

Annual training volume characteristics including accumulated hours ($\Delta = 201 \text{ h}$, $p \leq .001$, $d = 2.46$), covered distance ($\Delta = 6,849 \text{ km}$, $p \leq .001$, $d = 3.0$), climbing metres ($\Delta = 76,555 \text{ m}$, $p \leq .001$, $d = 1.36$), total sessions ($\Delta = 77 \text{ sessions}$, $p \leq .001$, $d = 2.21$) and total work ($\Delta = 168,104 \text{ kJ}$, $p \leq .001$, $d = .90$) were higher in U23_{ELITE} compared to U23_{NON-ELITE}.

Training and racing data revealed that total time and %Time in HR zone 1 ($\Delta = 182 \text{ h}$, 8.7% ; $p \leq .001$, 0.07 ; $d = 2.93$, 1.35 ; respectively) were higher in U23_{ELITE}, while only %Time in HR zone 2 ($\Delta = 2.5\%$, $p = .035$, $d = 1.02$) and zone 3 ($\Delta = 6.1\%$, $p = .007$, $d = 1.35$) were lower in U23_{ELITE} – see, Figure 2A,B. Total time in HR zone 2 ($\Delta = 14 \text{ h}$, $p = .140$, $d = .69$) and zone 3 ($\Delta = 5 \text{ h}$, $p = .805$, $d = .11$) were not different between groups. The intensity distribution displayed as total time in power output zone 1 ($\Delta = 138 \text{ h}$, $p = .003$, $d = 1.51$) and zone 3 ($\Delta = 15 \text{ h}$, $p = .011$, $d = 1.26$) were higher in U23_{ELITE}. %Time in power output zone 1 ($\Delta = 4.8\%$, $p = .257$, $d = .52$), total time and % Time in power output zone 2 ($\Delta = 50 \text{ h}$, 5.1% ; $p = .223$; $.076$; $d = .56$, $.84$) and %Time in power output zone 3 ($\Delta = 0.4\%$, $p = .525$, $d = .29$) were not different between groups – see, Figure

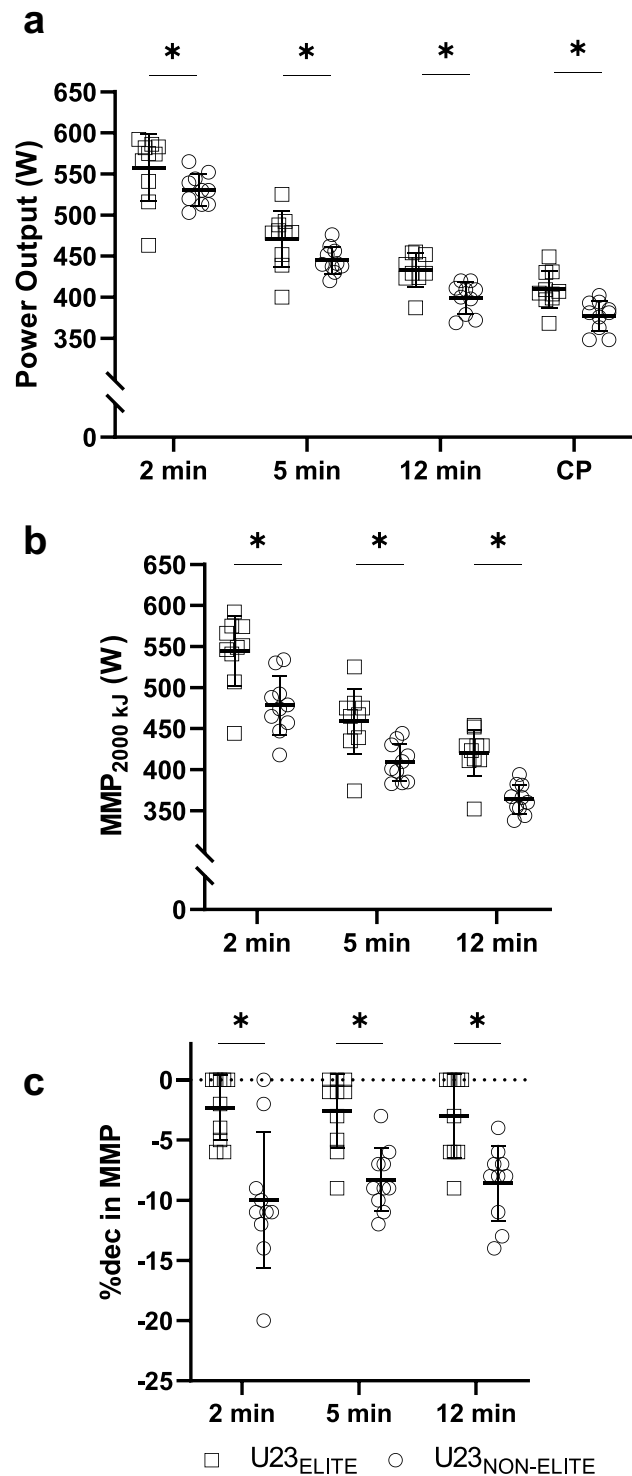


Figure 1. Power profiling field test (A), MMP after 2000 kJ (B) and the %dec in 2, 5 and 12 min power output after 2000 kJ (C) in U23_{ELITE} and U23_{NON-ELITE}; CP – critical power; MMP – mean maximum power; %dec – percentage decline; *significantly different between groups.

2C,D. Polarization index ($\Delta = .03 \text{ A.U.}$, $p = .334$, $d = .44$) and luTRIMP ($\Delta = 13,194 \text{ A.U.}$, $p = .086$, $d = .81$) showed no differences between groups.

No differences were found in race days ($\Delta = 7$, $p = .263$, $d = .51$) and race kilometres ($\Delta = 1,202 \text{ km}$, $p = .077$, $d = .84$) between groups. The UCI points scored were significantly

Table 2. Power-duration parameter estimates between U23_{ELITE} and U23_{NON-ELITE}; MMP – mean maximum power output; CP – critical power; W' – work above critical power; SE – standard error, RMSE – root mean square error; *significantly different from U23_{ELITE}.

	2-min MMP (W)	5-min MMP (W)	12-min MMP (W)	CP (W)	W' (kJ)	SE CP (W/%)	SE W' (kJ/%)	RMSE (W)
U23 _{ELITE} (n = 10)	557 ± 41 7.8 ± 0.4	471 ± 34 6.6 ± 0.3	433 ± 20 6.1 ± 0.4	410 ± 22 5.8 ± 0.4	17.8 ± 3.7	8 ± 8 2.0 ± 0.4	1.5 ± 1.5 8.4 ± 0.4	7.8 ± 7.8
U23 _{NON-ELITE} (n = 10)	530 ± 19* 7.9 ± 0.6	445 ± 16* 6.6 ± 0.4	398 ± 19* 5.9 ± 0.5	377 ± 19* 5.6 ± 0.4	18.6 ± 2.6	12 ± 8 3.1 ± 0.5	2.3 ± 1.5 12.3 ± 0.6	11.9 ± 8.0

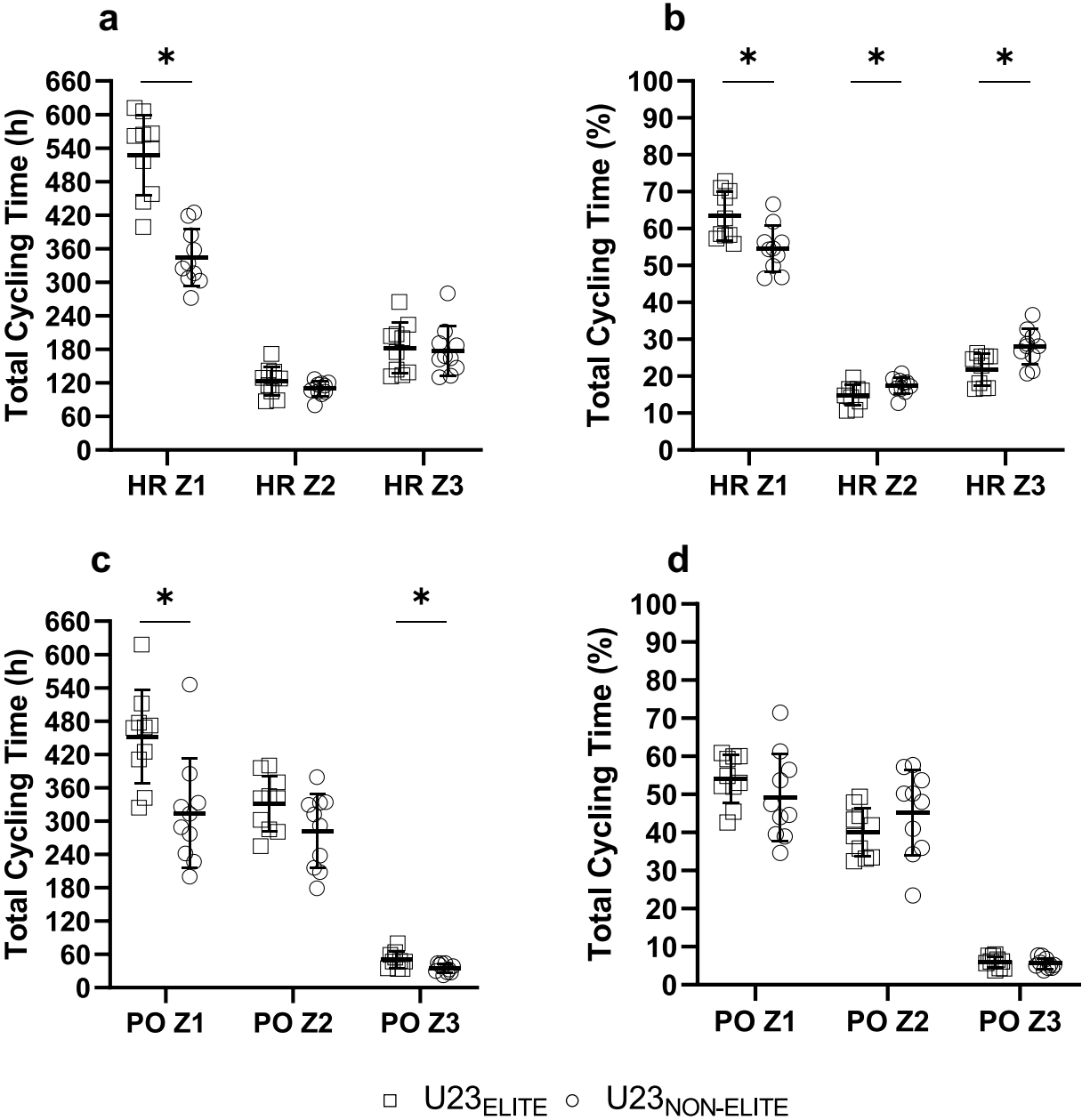


Figure 2. Training and racing intensity distribution based on Total Time (min) and %Time spent in heart rate (A and B) and power output (C and D) zones 1 to 3; HR – heart rate; PO – power output; *significantly different between groups.

higher in U23_{ELITE} ($\Delta = 300$ points A.U., $p \leq .001$, $d = 2.60$) compared to U23_{NON-ELITE}.

Key performance indicators

Certain key performance indicators of the power profile as well as training volume and intensity characteristics showed significant relationship with the UCI points scored – see, Table 3.

Discussion

The findings of the present study with U23 cyclists revealed: 1) that U23_{ELITE} are younger, 2) a higher absolute PPO and power profile in U23_{ELITE}, and 3) higher training volume characteristics in U23_{ELITE} compared to U23_{NON-ELITE} cyclists. Based on these main findings, both laboratory and field derived performance characteristics were identified as key performance indicators separating U23_{ELITE} and U23_{NON-ELITE} cyclists. In addition, the U23_{ELITE} group was by 0.8 years younger than the U23_{NON-ELITE}, which can be interpreted as U23_{ELITE} cyclist stepping up to the professional level earlier. To the authors' knowledge this is the first study investigating the physiological, performance and training/racing characteristics associated with transitioning from U23 cycling to the elite/international level.

Laboratory testing

Previous studies (Bell et al., 2017; Coyle, 1995; Hawley & Noakes, 1992; Lamberts et al., 2012) in elite level cyclists reported a strong relationship between laboratory assessed parameters (PPO, VO_{2max} , $\%VO_{2max}$) and cycling endurance performance. In contrast to our initial hypothesis, laboratory testing did only reveal significant differences for absolute PPO between U23_{ELITE} (486 ± 38 W) and U23_{NON-ELITE} (443 ± 33 W), while differences

in other laboratory parameters including relative and absolute power output at VT and RCP, and VO_{2max} did not reach statistical significance. However, despite not reaching statistical significance all laboratory metrics expressed in absolute terms showed moderate to large effect sizes ($d = .40$ to 1.19), indicating that these differences may be physiologically relevant. Consequently, it may be suggested that while laboratory metrics expressed in relative terms are obviously important performance characteristics for U23 cyclists, these metrics may not be considered as key performance indicators separating future performers from non-performers. Instead, it seems that these laboratory metrics expressed in absolute terms have the potential to discriminate between U23_{ELITE} and U23_{NON-ELITE} cyclists. Due to the small variation in body mass between groups laboratory characteristics (PPO, VO_{2max} , RCP and GET) when normalized to body mass were not statistically different. These findings are partially in line with Svendsen et al. (2018) who reported that absolute and relative PPO is a discriminator between future professional and non-professional cyclists. However, the relative PPO of future professionals reported in this investigation (6.9 W.kg^{-1}) is fully in line with the PPO observed in both groups herein. Group mean relative VO_{2max} values (U23_{ELITE}: 76.2 ml.kg^{-1} ; U23_{NON-ELITE}: 75.1 ml.kg^{-1}) were lower than previously reported by Zapico et al. (2007; 80.5 ml.kg^{-1}), but in line with Sallet et al. (2006; Elite: 73.9 ; U23: 75.7 ml.kg^{-1}). VO_{2max} and PPO values in the present study were higher than previously reported in high level junior and U23 cyclists (Menaspà et al., 2012) – VO_{2max} : 67.3 ml.kg^{-1} vs. 75.6 ml.kg^{-1} ; PPO: 6.2 vs. 6.7 W.kg^{-1} , respectively. These findings indicate that VO_{2max} and PPO are important determinant factors when comparing heterogeneous cycling groups (e.g., junior vs. U23; Menaspà et al., 2012; Svendsen et al., 2018), their predictive ability in more homogeneous groups (e.g., U23 vs. professional) is compromised. In addition a comparison of VO_{2max} and PPO between different studies needs to be interpreted cautiously due to different laboratory protocols and measurement techniques (Lamberts et al., 2009; Luttikholt & Jones, 2022).

Power profiling

The power profiling revealed a higher absolute 2, 5 and 12 min MMP, and CP in U23_{ELITE} cyclists, while no differences between groups were observed in W' . Interestingly and similar to the laboratory derived metrics, performance characteristics revealed by the power profiling and expressed relative to each cyclist's body mass were not significantly different between U23_{ELITE} and U23_{NON-ELITE} cyclists. Leo et al. (2021; 2020) demonstrated that the power profile during a competitive season is best characterized by changes in relative power output due to a progressive reduction in body mass from pre- to late-season periods, possibly stimulated by the accumulation of higher training loads and more racing days. In the present study, the relative power profile was not sensitive enough to differentiate between U23_{ELITE} and U23_{NON-ELITE} cyclists. For this reason, previous research proposed allometric scaling of body mass using mass exponents of 0.79 to 0.91 to compensate for the influence of air resistance on frontal drag (i.e. energy cost) between level ground and uphill cycling terrain (Jobson et al., 2008; Nevill et al., 2006; Swain, 1994). The absolute power profile demonstrated no statistical differences between U23

Table 3. Spearman's rank correlation coefficient between UCI points scored and key performance indicators; effect size for the correlation coefficient (r): ≤ 0.10 – small; ≤ 0.3 – medium; ≥ 0.8 large; %Time HR Z1 – Z3 – percentage training and racing time in heart rate zones 1 to 3; PO – power output; MMP – mean maximum power; CP – critical power; %dec – percentage decline; PPO – peak power output; *significant correlation.

	UCI points scored rho (ρ)	UCI points scored p-value
Hours	.673	.001*
Distance	.713	$\leq .001$ *
Climbing metres	.474	.035*
Total sessions	.638	.002*
Total work	.650	.002*
Time in PO zone 1	.572	.008*
Time in PO zone 3	.487	.029*
Time in HR zone 1	.668	.001*
%Time in HR zone 1	.603	.005*
%Time in HR zone 2	-.565	.009*
%Time in HR zone 3	-.546	.013*
PPO	.370	.108
2 min MMP	.425	.061
5 min MMP	.443	.050
12 min MMP	.555	.011*
CP	.563	.010*
2 min MMP _{2000KJ}	.561	.010*
5 min MMP _{2000KJ}	.487	.029*
12 min MMP _{2000KJ}	.546	.013*
%dec 2 min MMP	-.617	.004*
%dec 5 min MMP	-.610	.004*
%dec 12 min MMP	-.455	.044*

and professional cyclists during multi-stage racing in mountainous terrain, but the relative power profile was significantly different between groups. Based on these findings a high absolute power profile is a prerequisite for elite cycling, but further improvements in elite cycling performance might be realized by optimizing body composition and the power to mass ratio (Giorgi et al., 2018; Leo, Spragg, Simon, Lawley et al., 2021; Padilla et al., 1999). Moreover, level ground and uphill cycling conditions as well as cadence selection influence the power profile in field conditions (Bouillod et al., 2017; Hovorka et al., 2022; Nimmerichter et al., 2012). While we are aware of a potential influence on the power-duration parameter estimates (CP and W'), the variation in cadence between the 2, 5 and 12 min prediction trials at the 5.5% uphill climb was considerably small, i.e. 83 vs. 86 rpm. For this reason the ecological validity of the power profile test in this study was not violated as the methodological approach followed previous guidelines and recommendations (Leo, Spragg, Podlogar et al., 2021; Muniz-Pumares et al., 2019).

Apart from “fresh” power profiles, recent research focussed on the power profile after accumulated total work, (Leo, Spragg, Simon, Mujika et al., 2021; Muriel, Mateo-March et al., 2021; Muriel, Valenzuela et al., 2021; Spragg et al., 2022; Van Erp, Sanders et al., 2021) often referred to as fatigue resistance or durability (Mauder et al., 2021). In the current investigation, the power profile after 2000 kJ of accumulated total work derived during racing was used as an indicator of durability in U23 cyclists, as this was shown to discriminate between U23 and elite/international cyclists (Gallo et al., 2022; Leo, Spragg, Simon, Mujika et al., 2021). Both absolute and relative 2, 5 and 12 min MMP values after 2000 kJ of total work were lower in the U23_{NON-ELITE} group. This observation is of particular interest, especially considering the results of the laboratory and formal field tests presented herein. While a difference in absolute MMP values after 2000 kJ of total work between U23_{ELITE} and U23_{NON-ELITE} cyclists is obvious (as indicated by the difference in absolute MMP values derived from formal testing), the difference in relative MMP values between both groups indicates a superior fatigue resistance in the U23_{ELITE} cyclists. This notion is strengthened by the observation that the percentage decline in 2, 5 and 12 min MMP between the formal field test and the respective MMP values after 2000 kJ was on average 5.6 to 7.7% higher in the U23_{NON-ELITE} group. Previous research also reported that the decline in power output after accumulated work was smaller in world tour level compared to pro tour level cyclists during a cycling grand tour (Muriel, Valenzuela et al., 2021). Moreover, rider type classification between *allrounders*, *domestiques* and *general classification contenders* was also based on the decline in the power profile (Leo, Spragg, Simon, Mujika et al., 2021). Although this approach might facilitate the quantification of race demands and requirements, these findings need to be interpreted cautiously due to a major influence of team tactics and race strategies in professional road cycling (Ouvrard et al., 2018; Van Druenen & Blocken, 2021; Van Erp, Kittel et al., 2021). Despite these methodological considerations, the quantification of “durability” as a key performance indicator in highly trained

road cyclists is recommended, as values from formal power profile testing alone are not sufficient to differentiate between the U23 category and elite or world-class levels (Galloet al., 2022)

Training and racing characteristics

Training volume characteristics including accumulated hours, covered distance, climbing metres and total work were on average 31.8 to 38.2% higher in U23_{ELITE}. Intensity distribution based on total time and %Time in HR zones 1–3 showed more %Time in HR zone 1 (63.3% vs. 54.6%) for U23_{ELITE}, while U23_{NON-ELITE} spent more %Time in HR zone 2 (17.4% vs. 14.8%) and 3 (27.9% vs. 21.8%). In contrast, intensity distribution based on total time in power output zone 1 and 3 was higher in the U23_{ELITE} group, while %Time in power output zone 1 to 3 was not different between groups. Power output zones were corrected accordingly for the MRT (12 ± 1 W and 28 ± 3 s) in VO₂ between the laboratory GXT and constant work rate exercise (Boone et al., 2008; Iannetta et al., 2019). These findings are in line with previous research, which reported a U-shaped pattern in HR intensity distribution (Rosenblat et al., 2019; Stöggl & Sperlich, 2015; Zapico et al., 2007), while power output intensity distribution followed a pyramidal shape. Although the U23_{ELITE} group recorded more total time and %Time in HR zone 1, they also spent more total time in power output zone 3. These findings underpin that despite more accumulated time at higher power output, internally U23_{ELITE} demonstrated less %Time in HR zone 3 than U23_{NON-ELITE}. A similar trend was observed between professional and U23 cyclists during a multistage race, where professional riders recorded more accumulated time at higher power outputs and less accumulated time at higher HR intensities than the U23 group (Leo, Spragg, Simon, Mujika et al., 2021). One underlying mechanism for the discrepancy between external and internal time in zone might be improved fitness. Better trained athletes can spend more time at higher power outputs, without accumulating more internal work. This “aerobic decoupling” mechanism of internal and external workloads has been shown recently by marathon runners, where faster marathon finishers demonstrated less aerobic decoupling than slower finishers (Smyth et al., 2022). While both groups on average displayed a “non-polarized” polarization index (Treff et al., 2019), one third (N = 6) of the study participants followed a “polarized” training approach. There has been a recent debate on whether or not polarized training is optimal for endurance performance (Burnley et al., 2022; Foster et al., 2022). Recent research demonstrated that the training process often combines pyramidal and polarized intensity distribution approaches to maximize performance outcomes (Filipas et al., 2021). In addition, there was no separation between training and racing regarding intensity distribution, thus racing might have added more time in HR zone 2 as shown by previous research (Leo, Spragg, Simon, Mujika et al., 2021; Padilla et al., 2001). Given the purpose of the present study, only the entire season was analysed, so it was not possible to distinguish if the participants followed pyramidal or polarized training at different times within the season.

Racing characteristics showed higher UCI point scores in U23_{ELITE}, which can basically be interpreted as the U23_{ELITE} group being more successful in races. These findings are in accordance with previous research demonstrating that race success is an important determinant for a future elite/international cycling career (Gallo, Mostaert et al., 2022; Mostaert et al., 2021; Schumacher et al., 2006). However, no differences were found in the number of race days and race kilometres, indicating that there was no selection bias and equality and fairness were assured within the cycling team.

Key performance indicators

As shown in previous research (Leo, Spragg, Simon, Mujika et al., 2021; Muriel, Mateo-March et al., 2021; Van Erp, Sanders et al., 2021), laboratory derived parameters of aerobic fitness expressed in absolute terms, the power profile expressed in absolute terms, and “durability”, defined as the decline in MMP after accumulated total work, can be considered as key performance indicators discriminating between U23_{ELITE} and U23_{NON-ELITE} cyclists. Investigating the relationship between key performance indicators and race performance (UCI points scored) allows to determine the most impactful parameters. Based on our findings all training volume parameters showed a positive relationship with large effects ($p = .474$ to $.713$) on scored UCI points. This means that the higher the training volume, the higher the chance for scoring more UCI points. Interestingly, training intensity characteristics showed a biased trend to UCI points. While more total time and %Time in HR zone 1 was positively correlated to more UCI points, %Time in HR zone 2 and 3 negatively correlated ($p = -.546$ to $-.565$) with UCI points. More total time in PO zone 1 and 3 was also positively correlated to more UCI points. In other words, the more intense the participants trained internally by accumulating more total time and %Time in HR zone 2 and 3, the less UCI points they scored. This kind of “durability” marker for aerobic decoupling was previously shown by cyclists and runners as well (Leo, Spragg, Simon, Mujika et al., 2021; Smyth et al., 2022). Nonetheless it could also mean that the less fit riders accumulated more % Time in HR zone 2 and 3 during racing, which could impair the power profile and therefore reduce the chance for scoring more UCI points. This was confirmed by Leo et al. (2021; 2021) where U23 cyclists accumulated higher internal work than professional riders for the same external work load during multi-stage racing. Since intensity distribution involved both training and racing data, we can only speculate about the potential consequences.

The power profile positively correlated to UCI points with moderate to large effects ($p = .425$ to $.563$). In contrast a negative relationship with moderate to large effects ($p = -.455$ to $-.617$) was found between the decline in the power profile and UCI points. In other words, the more the cyclists declined in the power profile the less UCI points they scored. These findings are in line with previous research (Leo, Spragg, Simon, Mujika et al., 2021; Van Erp, Sanders et al., 2021), which demonstrated that changes in the power profile in combination with the decline after accumulated work and intensity is a key determinant for success in professional road cycling. However, these findings need to be interpreted with caution, as

work from Gallo et al. (Gallo et al., 2022) suggests that a high “fresh” power profile in junior level cyclists should be prioritised first, before further developing “durability” in U23 and elite level category. PPO from the laboratory as well as 2 and 5 min MMP derived from formal power profiling did not significantly correlate to UCI points. Therefore, characterizing rider calibre only based on laboratory and field test data might not be sufficient to determine a cyclist’s true performance capacity.

Limitations

The present study is not without limitations. Due to the small sample size the statistical power was reduced ($1 - \beta = 34.2$ to 49.4%), therefore the findings need to be interpreted with caution. A larger sample size would have been preferable, but 7 riders of the initial 27 did not meet the inclusion criteria of the study due to termination of cycling career or injury. Moreover, the high inter-individual variability in the performance measures of the U23_{ELITE} group might explain the large effect sizes, which underpins that there are no pre-favoured “stereotypes” of future professional cyclists. In addition, only the final season before turning professional or the 4th season of the U23 category was analysed for each rider, while it takes multiple years of training to reach an elite/international performance level (Pinot & Grappe, 2015). Regarding laboratory testing, a combination of step and ramp incremental GXT would have been preferable to be able to quantify gross efficiency, as this has been shown to be a relevant endurance performance parameter (Coyle, 1999; Jobson et al., 2012). Additionally the limitation of using fixed duration MMP values after accumulated total work has been discussed in previous research (Leo, Spragg, Podlogar et al., 2021), however up to date it represents the only valid approach to evaluate the decline in the power profile during competition (Gallo et al., 2022; Leo, Spragg, Simon, Mujika et al., 2021; Van Erp, Sanders et al., 2021).

Conclusion and future research directions

This study revealed that U23_{ELITE} and U23_{NON-ELITE} cyclists can be differentiated by their absolute power profile, as well as training and racing characteristics. Based on these findings, it is recommended that the professional cycling ecosystem including team, managers and performance directors incorporate a multi-faceted approach to cyclists’ physiological, performance and talent identification combining testing, monitoring, and observation. Future research is recommended for tackling the “fatigue resistance” or “durability” objectives regarding talent identification, testing protocols, performance benchmarking and the underlying physiology.

The present results demonstrate that the transition from U23 to the elite/international level requires not just a set of physiological and performance qualities, but also exceptional dedication and investment in the training process. In this context, future champions seem to be made, not born.

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